

National University of Computer and Emerging Sciences



Lab Exercise 08 AL2002-Artificial Intelligence Lab

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Exercise

1. Life Expectancy Dataset

- a. Load the dataset using `pandas.read_csv()`
- b. Display the first 5 rows
- c. Use `df.info()` and `df.describe()` to get a basic understanding
- d. Check how many missing values are present per column.
- e. Drop 'Population' and 'GDP' columns.
- f. For the remaining missing values:
 - Fill numerical columns using median.
 - Fill categorical columns (like "Status") using mode.
- g. Check the dataset size after dropping.
- h. Check if any duplicate rows exist
- i. Remove them using `drop_duplicates()`
- j. Check the dataset size after dropping.
- k. List all numerical columns
- l. Apply suitable feature scaling technique on each numerical column and mention why you choose it.
- m. List all the categorical columns
- n. Apply suitable feature encoding technique on each categorical column and mention why you choose it.
- o. Create a line plot showing how Life expectancy changes from 2011 to 2015.
- p. Create a grouped bar chart comparing the following variables across years:
 - infant deaths
 - under-five deaths
 - Adult Mortality
- q. Visualize a correlation heatmap for all numerical features, identify the features that are highly correlated with Life expectancy (based on absolute correlation values), determine the top 3 least correlated features with respect to Life expectancy, and drop those least correlated features from the dataset.
- r. Save the cleaned dataset as `cleaned.csv` and submit it with your `.ipynb` file

2. Consider a sample dataset below

```
data = {  
    'Age': ['25', '30', '35', '40', '45'],  
    'Income': ['$1000.5', '$1500.0', '$2000.75', '?', '$3000.25'],  
    'Signup_Date': ['2023-01-15', '2023-02-20', '2023-03-25',
```

```
'2023-04-30', '2023-05-05'],  
    'Color': ['Red', 'Blue', 'Green', 'Red', 'Blue'],  
    'Score': [85, 90, 78, 92, 88]  
}  
  
df = pd.DataFrame(data)  
df
```

Convert each of the given columns in a suitable datatype after cleaning. Display the data type of all columns before and after conversion